Matplotlib Seaborn Plotly Basics

matplotlib,seaborn and plotlybasics for ml using python compare theory and code side by side

If you're learning **Machine Learning with Python**, the three main visualization libraries are:

Library	Best For	Learning Difficulty	Interactivity	Used in ML
Matplotlib	Basic charts and customization	⭐⭐	❌ Static	⭐⭐⭐⭐⭐
Seaborn	Statistical visualization	⭐⭐⭐	❌ Static	⭐⭐⭐⭐⭐
Plotly	Interactive dashboards	⭐⭐⭐⭐	✅ Interactive	⭐⭐⭐⭐

Comparison Overview

Feature	Matplotlib	Seaborn	Plotly
Built on	Base plotting library	Matplotlib	JavaScript (Plotly.js)
Ease of Use	Medium	Easy	Easy
Default Style	Plain	Beautiful	Interactive
Statistical Charts	Manual	Built-in	Limited
Interactive	No	No	Yes
Dashboard Support	No	No	Yes
Industry Use	Very High	Very High	High

1. Line Plot

Theory

Matplotlib

Everything is created manually.

Seaborn

Automatically creates attractive statistical plots.

Plotly

Creates interactive plots with zoom and hover.

Code Comparison

```python
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import pandas as pd

df = pd.DataFrame({
    "Day":[1,2,3,4,5],
    "Sales":[100,150,130,180,210]
})
```

Matplotlib

```python
plt.plot(df["Day"], df["Sales"])
plt.title("Sales Trend")
plt.xlabel("Day")
plt.ylabel("Sales")
plt.show()
```

Seaborn

```python
sns.lineplot(data=df, x="Day", y="Sales")
plt.title("Sales Trend")
plt.show()
```

Plotly

```python
fig = px.line(df, x="Day", y="Sales", title="Sales Trend")
fig.show()
```

Output Comparison

Matplotlib	Seaborn	Plotly
Basic line	Styled line	Interactive line

---

2. Scatter Plot

Theory

Used to find relationships between variables.

Example:

- Experience vs Salary
- Age vs Income

Matplotlib

```python
plt.scatter(df["Day"], df["Sales"])
plt.show()
```

Seaborn

```python
sns.scatterplot(data=df, x="Day", y="Sales")
plt.show()
```

Plotly

```python
fig = px.scatter(df, x="Day", y="Sales")
fig.show()
```

3. Bar Chart

Theory

Compare categories.

Example:

Product	Sales
A	200
B	150

Matplotlib

```python
plt.bar(df["Day"], df["Sales"])
plt.show()
```

Seaborn

```python
sns.barplot(data=df, x="Day", y="Sales")
plt.show()
```

Plotly

```python
fig = px.bar(df, x="Day", y="Sales")
fig.show()
```

4. Histogram

Theory

Shows data distribution.

Used for:

- Age
- Salary
- Marks

Matplotlib

```python
plt.hist(df["Sales"])
plt.show()
```

Seaborn

```python
sns.histplot(df["Sales"])
plt.show()
```

Plotly

```python
fig = px.histogram(df, x="Sales")
fig.show()
```

5. Box Plot

Theory

Detects:

- Median
- Quartiles
- Outliers

Essential for data cleaning.

Matplotlib

```python
```

```
plt.boxplot(df["Sales"])
plt.show()
```

Seaborn

```python
sns.boxplot(y=df["Sales"])
plt.show()
```

Plotly

```python
fig = px.box(df, y="Sales")
fig.show()
```

6. Violin Plot

Theory

Shows:

- Distribution
- Density
- Median

Better than a box plot when you want to see the shape of the data.

Matplotlib

Not directly available.

Seaborn

```python
sns.violinplot(y=df["Sales"])
plt.show()
```

Plotly

```python
fig = px.violin(df, y="Sales")
fig.show()
```

7. Heatmap

Theory

Used to visualize correlation matrices.

Highly important in machine learning for feature analysis:

```Python
corr = df.corr(numeric_only=True)
```

Matplotlib

```Python
plt.imshow(corr)
plt.colorbar()
plt.show()
```

Seaborn

```Python
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()
```

Plotly

```Python
fig = px.imshow(corr, text_auto=True)
fig.show()
```

8. Pair Plot

Theory

Compares every feature with every other feature.

Commonly used during Exploratory Data Analysis (EDA).

Matplotlib

Requires manually creating multiple scatter plots.

Seaborn

```Python
sns.pairplot(df)
```

Plotly

No direct equivalent in Plotly Express (you can use `px.scatter_matrix`):

```Python
```

```
fig = px.scatter_matrix(df)
fig.show()
```

9. Correlation Matrix

```python
corr = df.corr(numeric_only=True)
```

Matplotlib

```python
plt.imshow(corr)
plt.show()
```

Seaborn

```python
sns.heatmap(corr, annot=True)
```

Plotly

```python
fig = px.imshow(corr)
fig.show()
```

10. Distribution Plot

Seaborn

```python
sns.kdeplot(df["Sales"], fill=True)
```

Plotly

```python
fig = px.histogram(df, x="Sales", marginal="rug")
```

Matplotlib does not have a single high-level function for this; you would typically combine `plt.hist()` with a density estimate from another library.

Typical ML Workflow

```
Dataset
   │
   ▼
Pandas
   │
   ▼
EDA
   │
```

```
    ├── Matplotlib
    ├── Seaborn
    └── Plotly
    │
    ▼
Feature Engineering
    │
    ▼
Model Training
    │
    ▼
Evaluation
```

Which Library Should You Use?

Task	Best Library	Why
Learning plotting fundamentals	Matplotlib	Foundation for many Python plotting libraries
Exploratory Data Analysis (EDA)	Seaborn	Attractive defaults and statistical visualizations
Correlation analysis	Seaborn	Excellent **`heatmap()`** support
Presentation and dashboards	Plotly	Interactive charts with zoom, pan, and hover
Model evaluation plots	Matplotlib + Seaborn	Most ML libraries integrate naturally with them
Web applications (e.g., Dash)	Plotly	Designed for interactive web visualizations

Recommendation for ML beginners

1. **Start with Matplotlib** to understand plotting fundamentals (**`figure`**, **`axes`**, labels, legends, and customization).
2. **Learn Seaborn** for fast, high-quality statistical visualizations used in EDA.
3. **Finish with Plotly** to build interactive visualizations and dashboards for sharing insights.

This progression matches how many data scientists work in practice: **Matplotlib** → **Seaborn** → **Plotly**.